When More Observation Hurts: Redundancy and Robust Clarification in Ask-or-Act

CS6208 Final Presentation | 5-min talk

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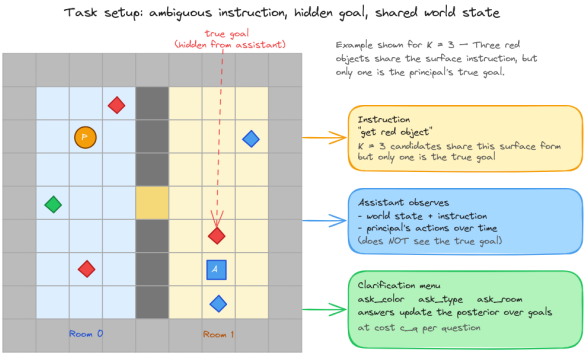
The Setup



Two ways to figure out a goal

Watch what the principal does, or
ask — each question costs
one turn + a small penalty.

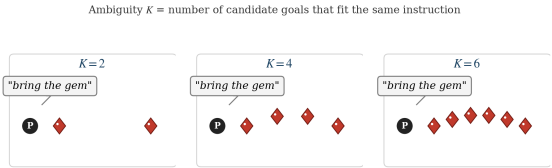
Three policies

NeverAsk — watch, infer, then commit.
AlwaysAsk — keep asking while unsure.
AskOrAct — compare ask cost vs. act cost and choose the cheaper one.

The question

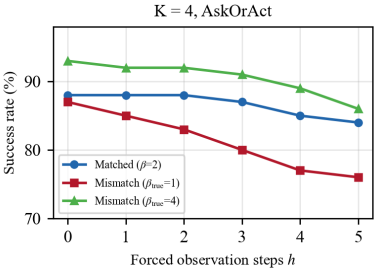
How do these channels interact —
and when does more **watching** *actually* **hurt***?*

Main Finding: Watching Can Hurt



Same instruction, K candidate goals.

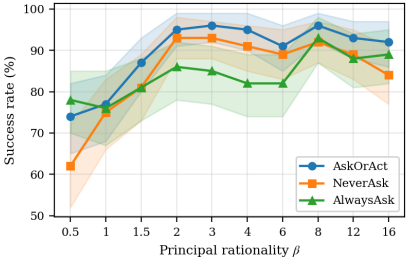
Higher $K \Rightarrow$ more objects match
— more to disambiguate.



When the model is wrong, watching makes things worse.

Every extra observation step pulls the posterior further astray.

A Surprise — and the Fix



A more rational partner is not always easier to help.

Their actions stop varying — so a smart assistant asks *more*.

The fix

When an observation looks surprising, **trust it less.**

Matched model: no harm.
Wrong model: recovers the loss.

Takeaways

1. Watching and asking *overlap* — don't stack them.
2. A more rational partner is not always easier.
3. When your model is wrong, fix *what you believe*, not *when you speak*.

Thank you

Questions?

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